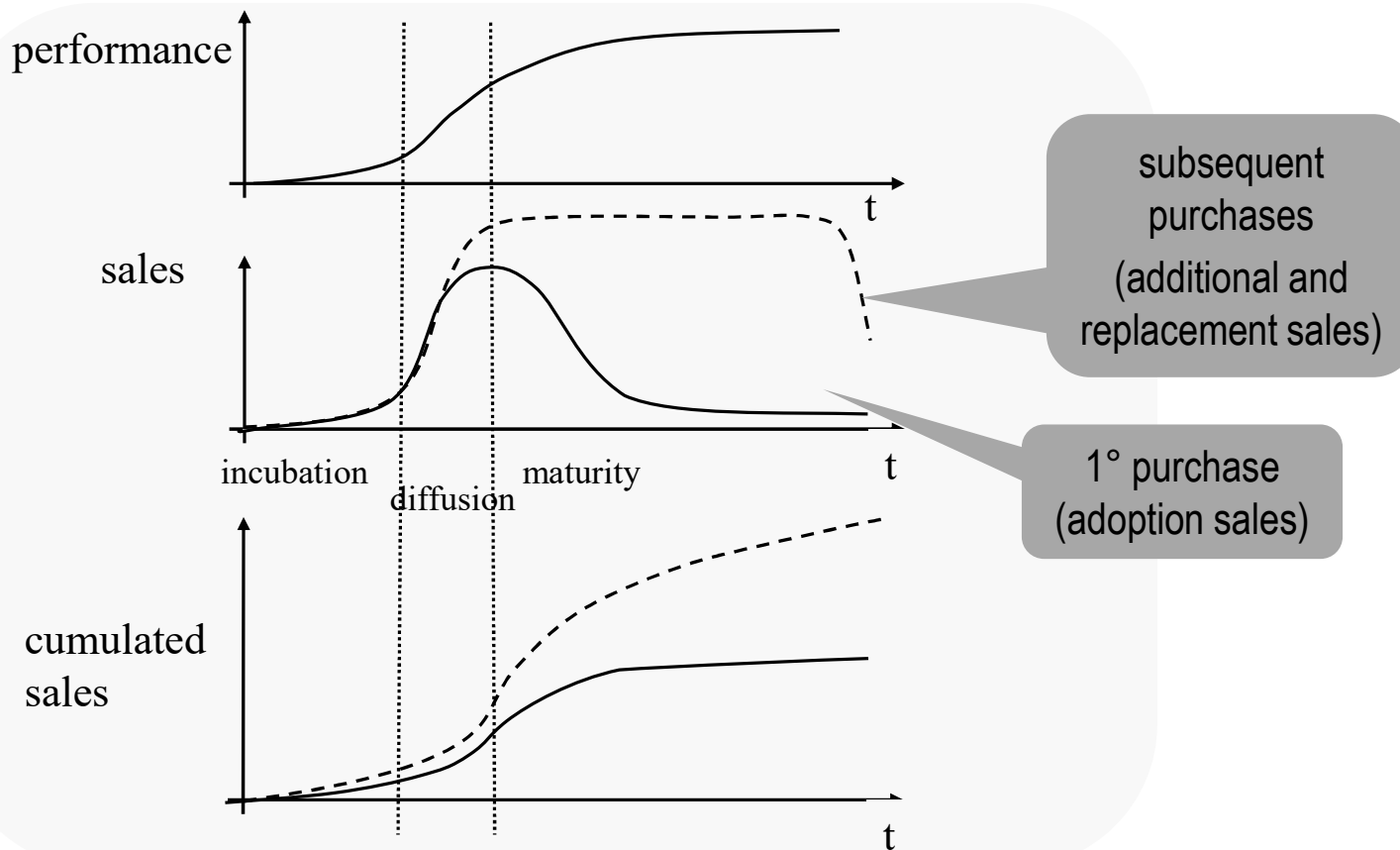

THE DYNAMICS OF INNOVATION

Dynamics of innovation

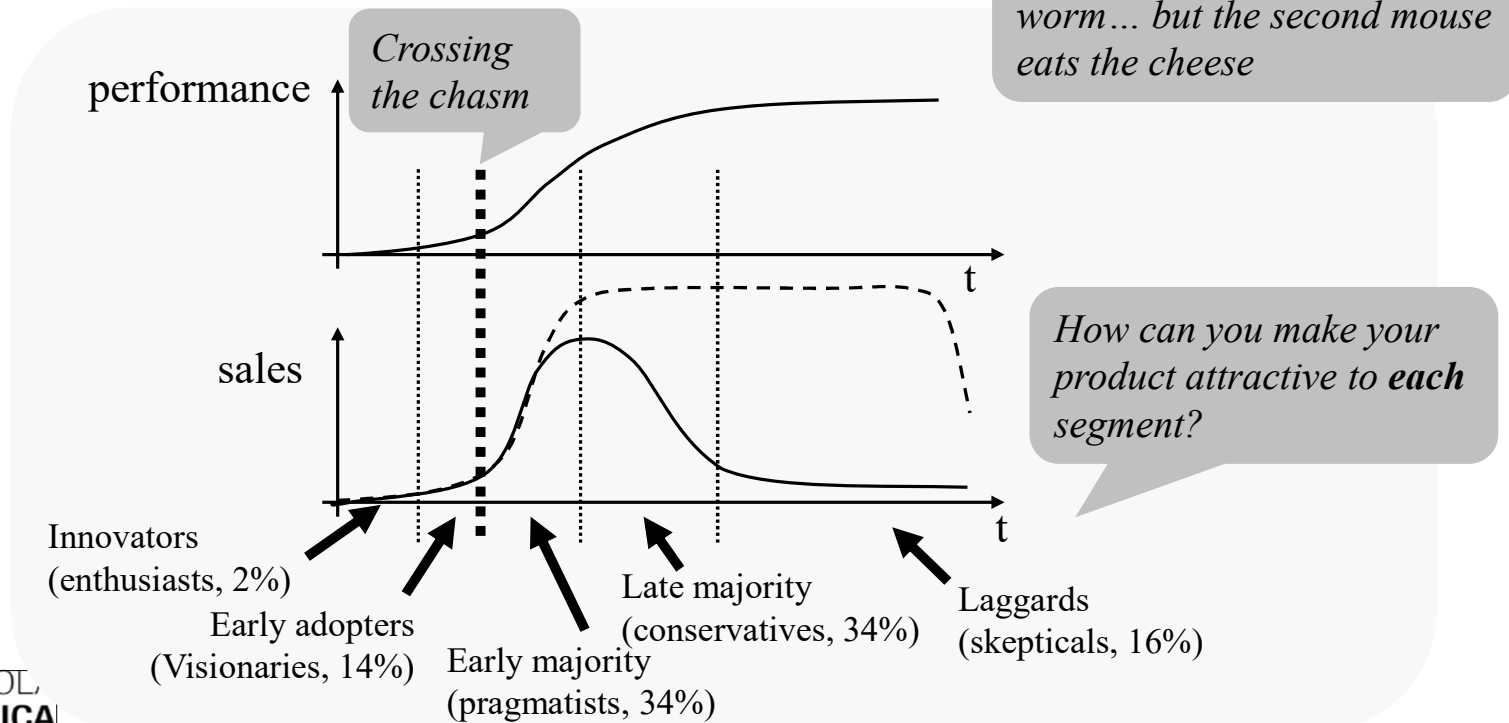
- Performance s-curves and diffusion s-curves



Dynamics of innovation

- Moore's segmentation

- Customers along the diffusion curve belong to distinct segments
- Moving to the early majority requires new competencies and assets
- Is it better to be leaders or followers? It depends....





Diamond Rio
PMP300, 1998, 32 MB

Apple iPod, 2001,
5 GB (still not
Windows
compatible...
Windows versions
come in 2002)



Archos Jukebox Recorder
15, 2001, 15 GB



Compaq-HanGo Personal
Jukebox, 1999, 5 GB



Apple iTunes,
2003

Dynamics of innovation

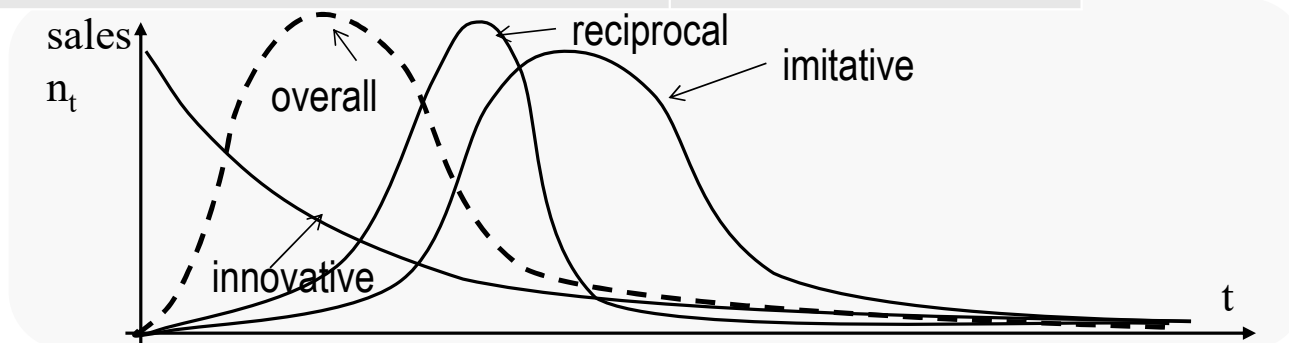
- The shape of diffusion curves (Bass model and subsequent)

Type of diffusion	What happens	In mathematical terms (n = sales; N = adopters; M = market potential)
Innovative	The producer advertises, and a fraction of non-adopters adopt	$n_t = p(M - N_t)$
Imitative	Adopters - generate "word of mouth" among non-adopters - create network externalities, which increase value to non-adopters (if standards-based) - generate margins, which are reinvested in sales effort	$n_t = q \frac{N_t}{M} (M - N_t)$
Imitative - Reciprocal	In the case of two-sided markets, adopters on side 1 attract adopters on side 2, and so on	$n_t = q \frac{N_t}{M} e^{\frac{N_t}{M}}$

This is called epidemic diffusion.

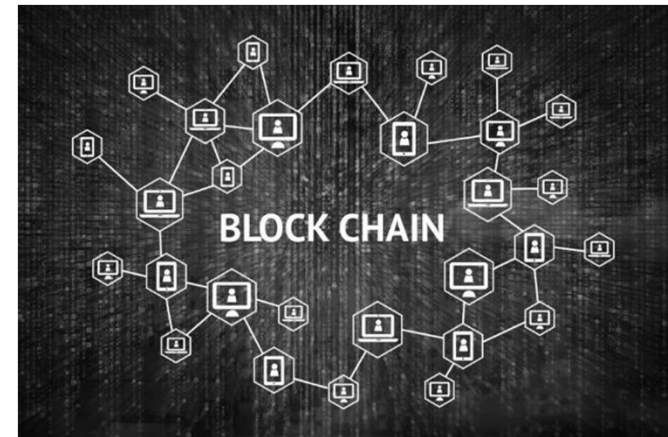
The main difference is that, in real epidemics, people do not remain "infectious" forever, but only for t_i days
The R_t parameter you read in the media is given by

$$R_t = q t_i$$



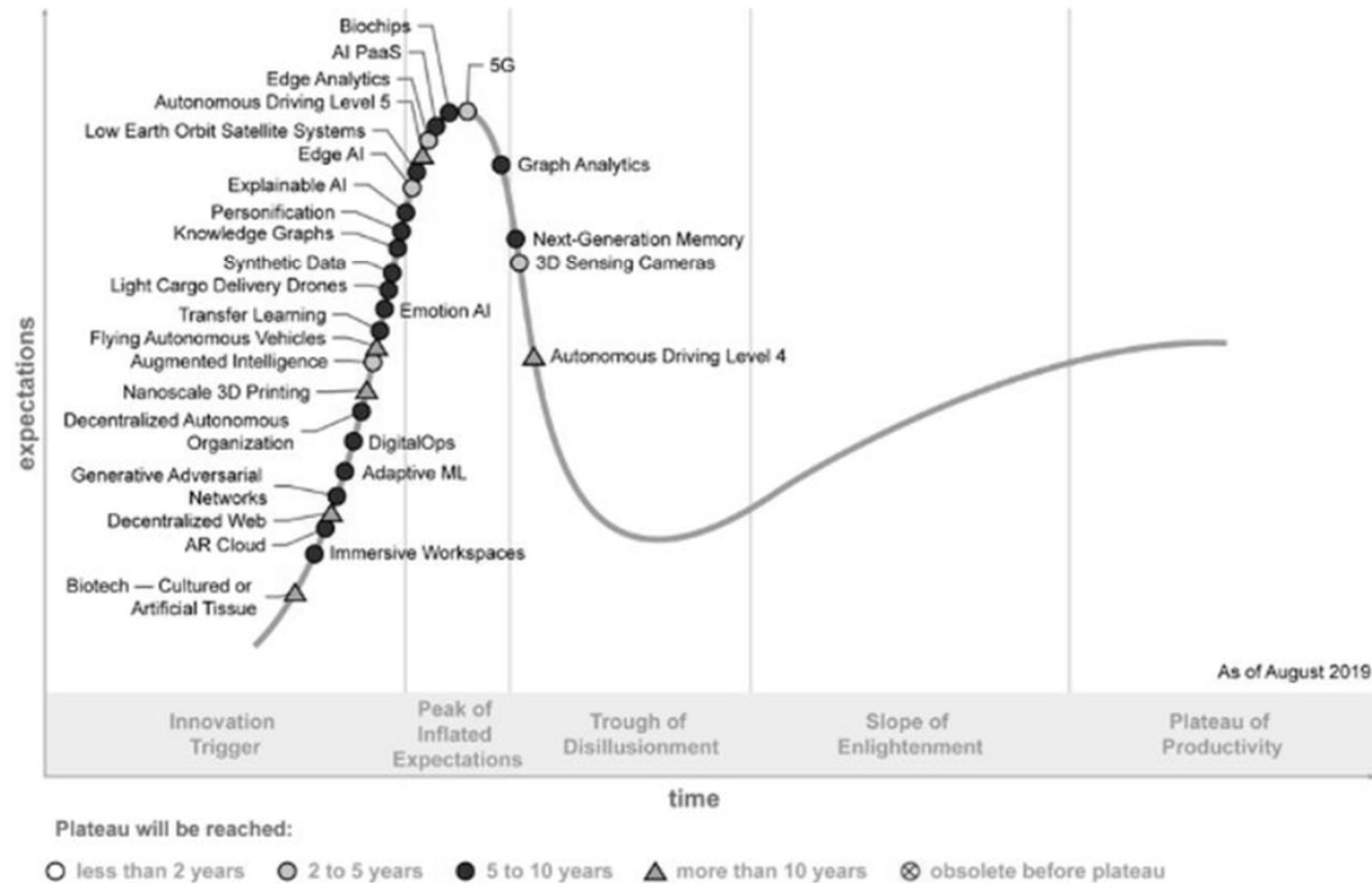
Dynamics of innovation

- Technology often suffers from *hype* and needs to mature before diffusing (trigger → incubation → diffusion)
- The incubation phase has a crucial role in
 - Improving technology in absolute terms
 - Finding applications that really matter (without unrealistic expectations)
 - Directing technology to these “real” applications
 - Raise awareness and confidence in customers
 - Reshuffling industry players
- The Gartner consultancy studies “hype cycles” for emerging technologies in the IT industry



Dynamics of innovation

Hype Cycle for Emerging Technologies, 2019

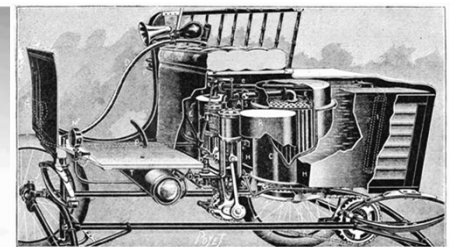
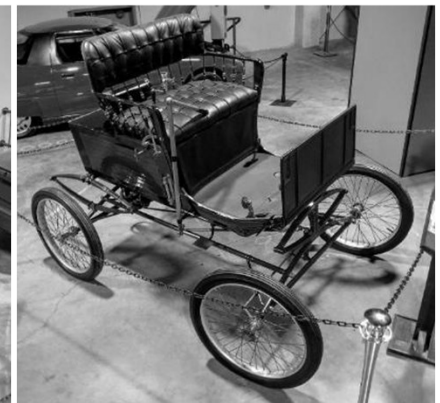


Source: Gartner
ID: 370466

Dynamics of innovation

- Connecting supply and demand (Abernathy-Utterback model for assembled products) Gallo et al., 2014

Cars in the XIX-XX Century
What powertrain?



Gasoline

Electric

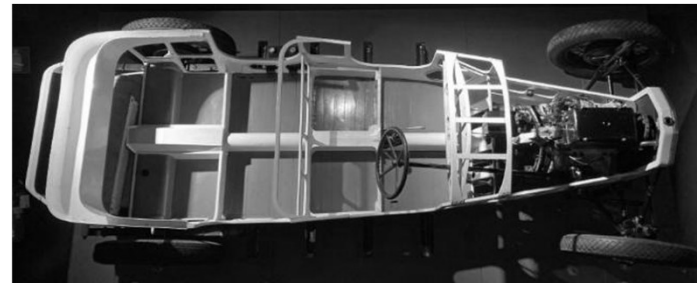
Steam

Dynamics of innovation

- Connecting supply and demand (Abernathy-Utterback model for assembled products) Gardner, 2014

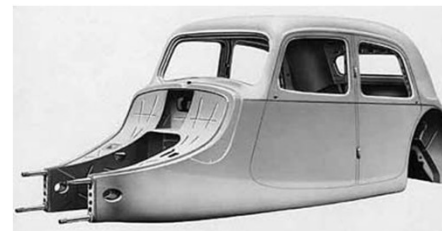
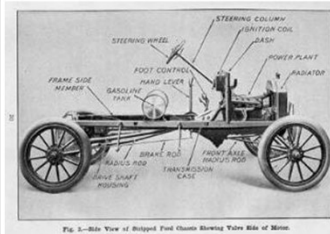
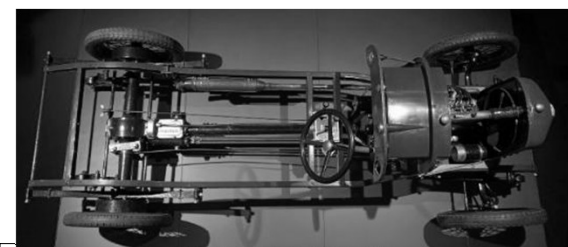
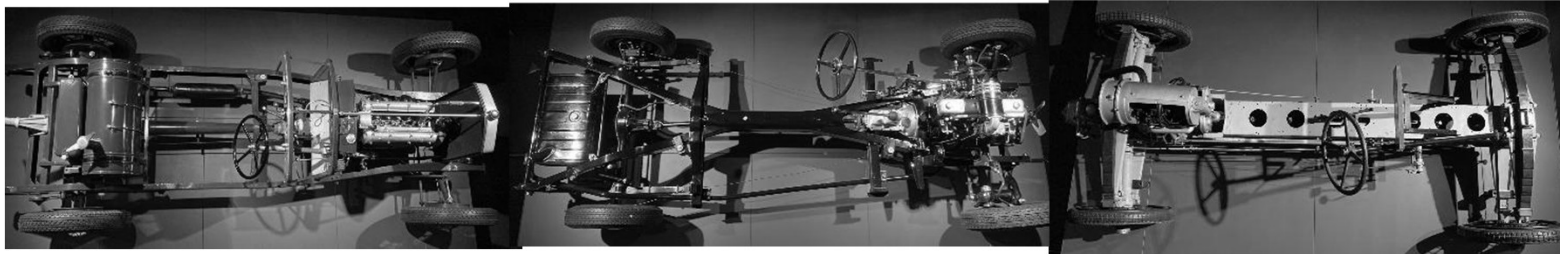
What body structure?

Buggy

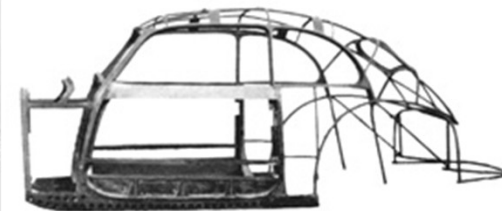


Boat hull, with keel and beams

Chassis with different geometries



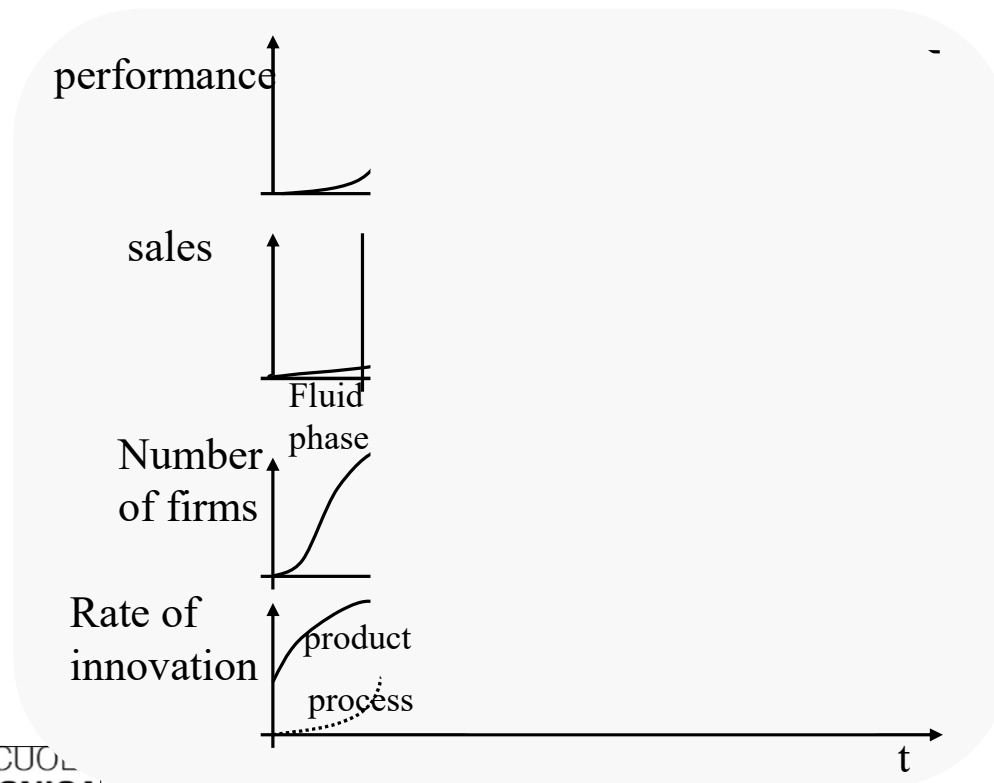
Unibody / monocoque



Space frame

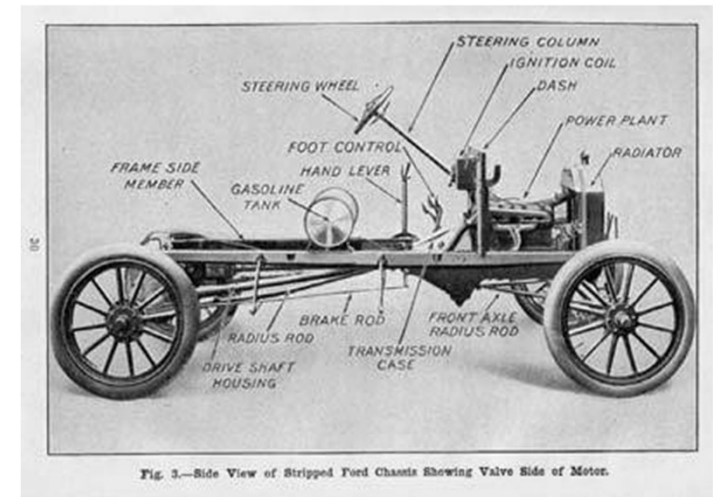
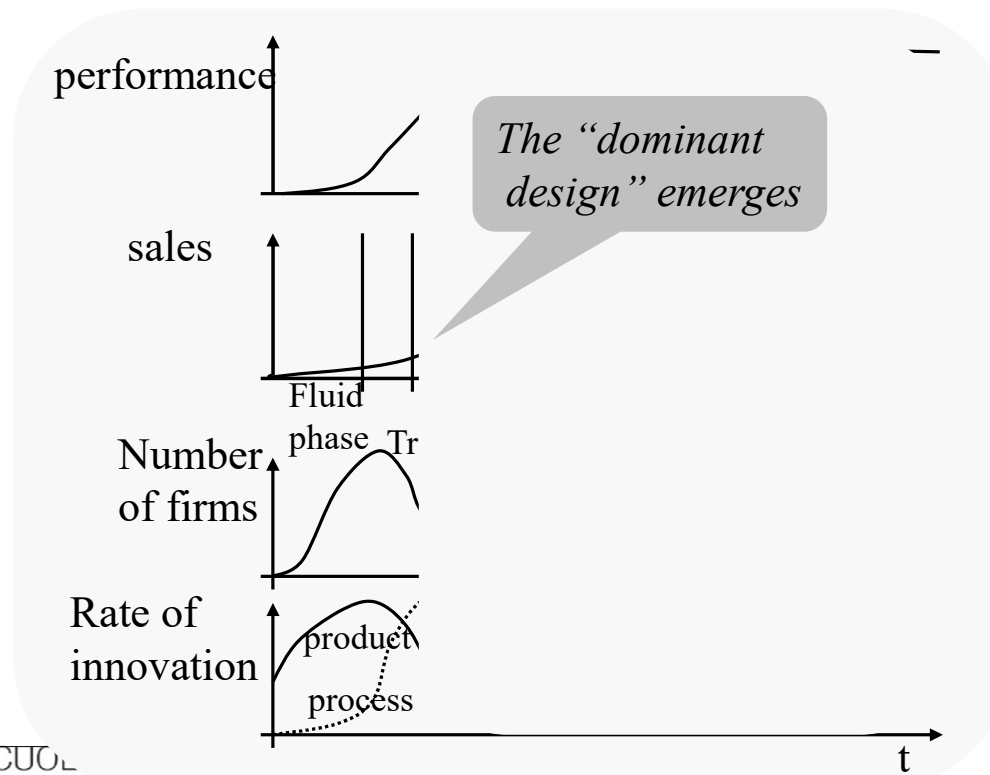
Dynamics of innovation

- Connecting supply and demand (Abernathy-Utterback model for assembled products)



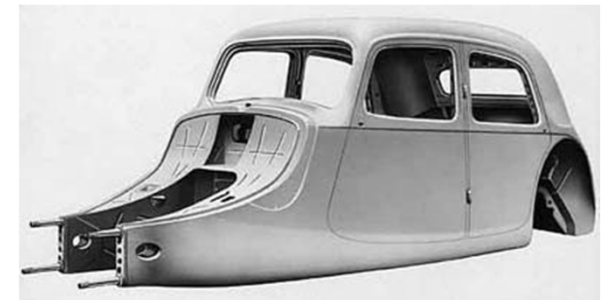
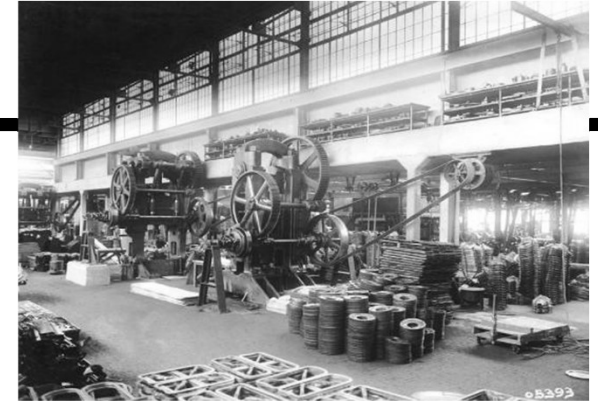
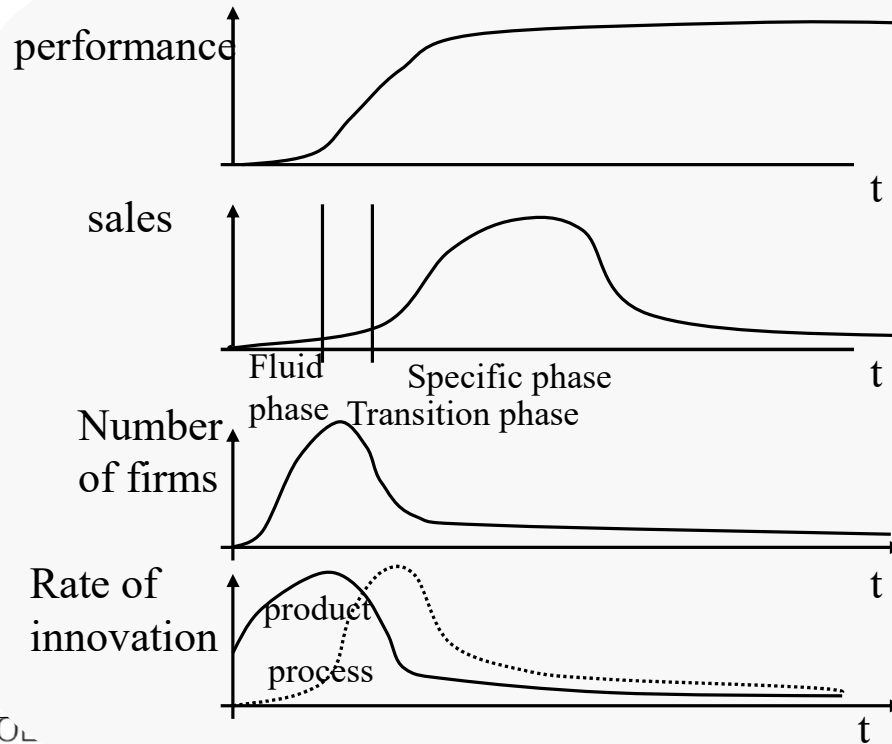
Dynamics of innovation

- Connecting supply and demand (Abernathy-Utterback model for assembled products)



Dynamics of innovation

- Connecting supply and demand (Abernathy-Utterback model for assembled products)

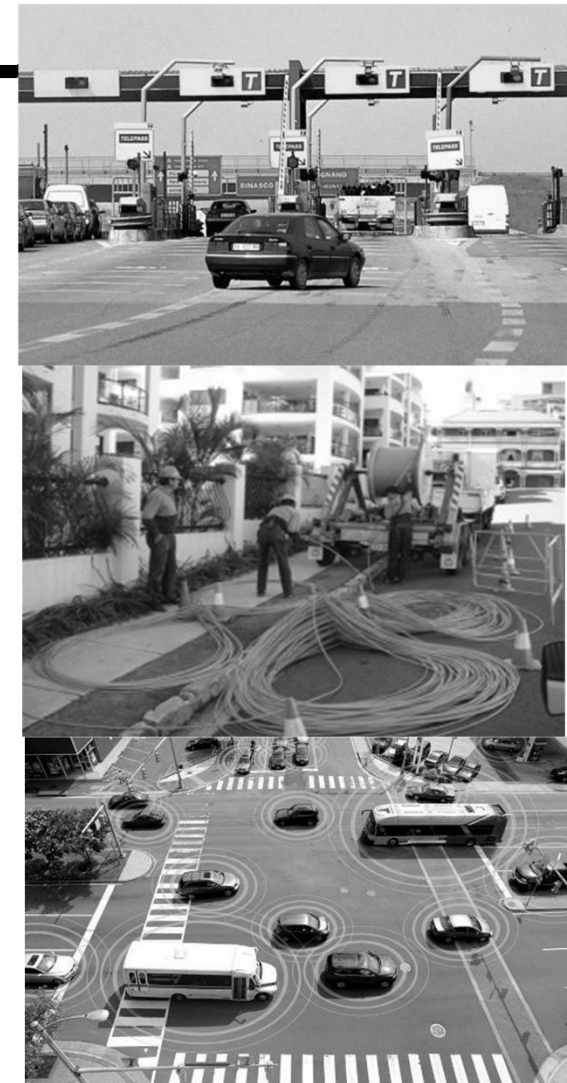


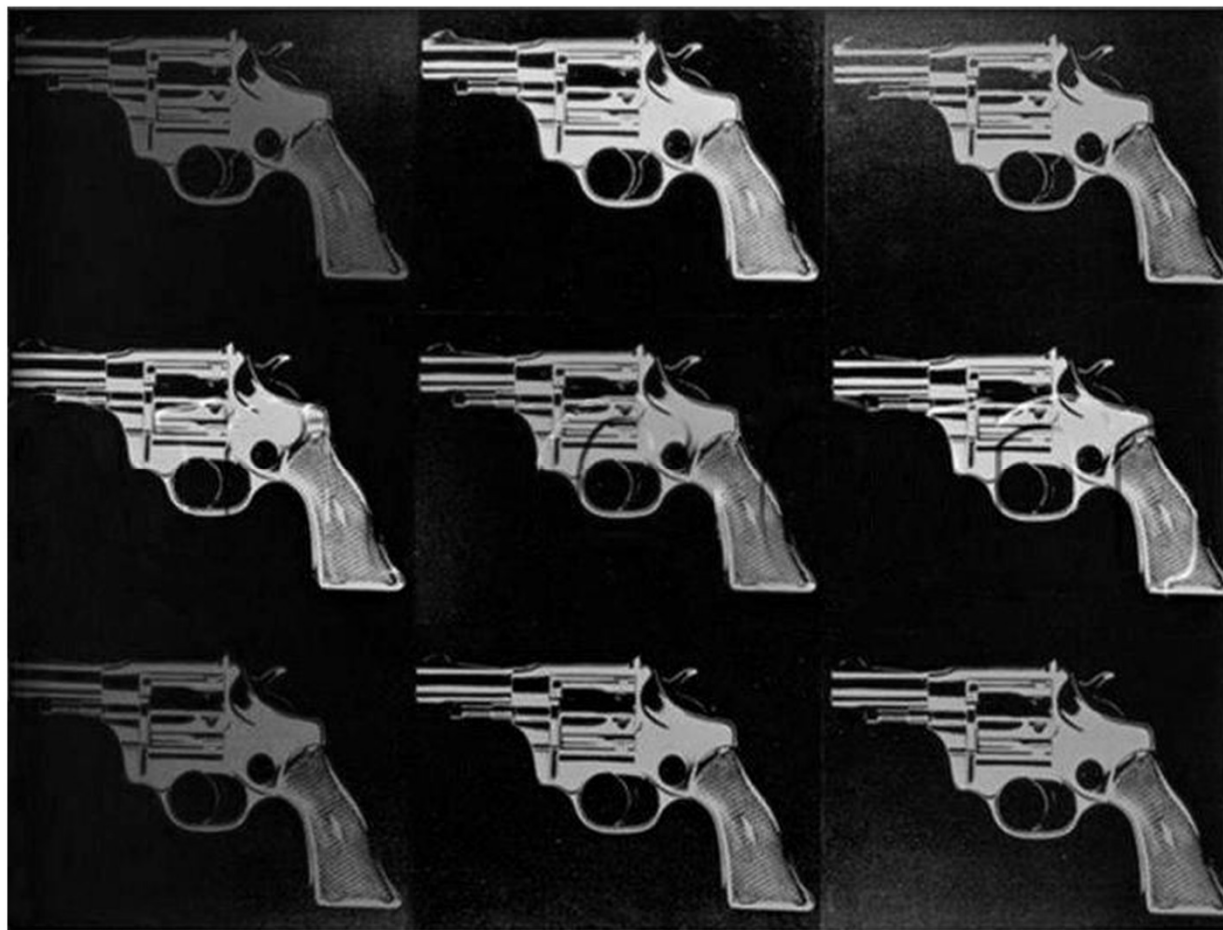
Dynamics of innovation

Questions	Possible answers
Why does a specific design become dominant over its competitors?	• It is perceived as superior (... by initial customer niches!) • Complementary assets are available (chicken & egg problem) • Diffusion occurs across many subsequent niches, and with little reaction from actors affected by disruption • Firm(s) following the design have a strong reputation, especially if they allow technology to spill over to competitors • Public policy • Product and process innovations are viable
What happens to unsuccessful designs?	• They disappear • They end up serving niches • They "feed" the successful design
What happens to the dominant design?	The industry gets locked in for a long time, because of • economies of scale • investment in specific complementary assets • organizational learning within firms and value chains

Dynamics of innovation

Questions	Possible answers
What happens in the case of the process industry?	• Innovation is difficult to observe (i.e., based on technology that can't be reverse engineered, and on organization) • Dominant designs emerge in the process (not in the product) • The temporal relationship between rates of innovation for the product and the process is inverted
And in the case of services?	• Same as in the case of the process industry
What are the consequences?	• The infrastructure dilemma: «Let's develop a cheap and basic process that is fine for today ... but which will create lock in and prevent future product innovation» vs. «Let's develop a high-end process technology that may open up future product innovation ... but is expensive, may nonetheless become obsolete, and for which all this wonderful product innovation might not materialize»





Standards

- Standard: *“set of specifications that provide value to the product because of its conformity to the standard”*
 - A standard is there if "were the product different, it would be of lesser economic value because of non-conformity (and NOT because of lesser performance!)"



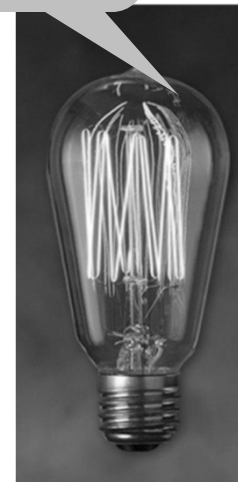
Voltage (V)
Rechargeability (yes / no)
Diameter (mm)
Height (mm)
Shape / interface
Capacity (mAh)
Weight (g)

Standards

- Standard: *“set of specifications that provide value to the product because of its conformity to the standard”*
 - A standard is there if "were the product different, it would be of lesser economic value because of non-conformity (and NOT because of lesser performance!)"
 - Standards lead to strong lock-in effects

Sources	Examples
Network externalities	Telephone, What'sApp, data exchange formats
Complementarity with other goods	HW/SW, operating systems/applications
Specific learning	User interfaces (keyboards, icon-based systems), maintenance procedures
Modularity	Bus architectures on PCs
Economies of scale	Screws, threads

Do you see standards here?



Standards

- Standard: “*set of specifications that provide value to the product because of its conformity to the standard*”
 - A standard is there if "were the product different, it would be of lesser economic value because of non-conformity (and NOT because of lesser performance!)"
 - Standards may (or may not be) dominant designs

	Dominant design	Non dominant design
Standard	BluRay DVDs	Specifications of the Edison screw QWERTY keyboard
Non standard	Monocoque automobiles	-

Dominant designs and standards

Fighting a standards war requires specific strategies

- Standards can arise
 - De facto (standards war → monopoly)
 - By agreement
 - De iure

Standards wars can be socially undesirable

- Proprietary standards lead to monopolies, often benefiting foreign firms
 - Adoption is delayed
 - Risk of being locked in a low-quality standard which just happened to come out sooner
- Government intervention (on what basis?)

Standards war dilemma

- Agreeing (*«let's make sure there is one big cake and then we can compete for a slice»*)
vs.
fighting (*«let's try to get 100% of the cake, with the risk of getting nothing... and a possibly a smaller cake»*)
- The decision is influenced by the number of competitors
- Technical tradeoffs that have different appeal to the parties might make agreement impossible